Motor Speed Controller

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RBT211: Embedded Programming

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Code

```c
/*
 * btrMotorPWMV1.c
 *
 * Author : Levi Terry
 * phase    motor strength
    0       100%
    1       50%
    2       75%
    3       25%
 */

#include <avr/io.h>
#include <avr/interrupt.h>

//define global variables
volatile uint8_t phase = 0;
volatile int ovrFloCt = 0;

// Start TIMER0
static inline void initTimer0(void){
    TCCR0B |= (1 << CS01 | 1 << CS00);
    TIMSK0 |= (1 << OCIE0B);
    TIMSK0 |= (1 << TOIE0);
    sei();
}

//AFTER TIMER HITS 255, RUNS OVF
ISR(TIMER0_OVF_vect){
    // Flip the enable
    PORTD |= (1 << 2);
    // Set PD6 LOW
    PORTD &= ~(1 << 6);
    // Set PD5 HIGH
    PORTD |= (1 << 5);
    ovrFloCt++;

    // Wait for 10000 overflows (~10s)
    if(ovrFloCt == 10000){
        // Phase 0: go to phase 1 and 50% power
        if (phase == 0){
            phase = 1;
            OCR0B = 128;
        // Phase 1: go to phase 2 and 75% power
        } else if (phase == 1){
            phase = 2;
            OCR0B = 191;
        // Phase 2: go to phase 3 and 25% power
        } else if (phase == 2){
            phase = 3;
            OCR0B = 64;
```

```c
        OCR0B = 64;
        // Phase 3: Go to phase 0 and 100% power
        } else if (phase == 3){
            phase = 0;
            OCR0B = 255;
        }

        // Reset overflow count
        ovrFloCt = 0;
    }

}

//WHEN TIMER HITS OCR0B VALUE
ISR(TIMER0_COMPB_vect){
    // Flip enable pin for PWM
    PORTD &= ~(1 << 2);
}


int main(void)
{
    // Set Pins PD2, PD5, and PD6 to outputs
    DDRD |= (1 << 2) | (1 << 5) | (1 << 6);

    // Start the timer
    initTimer0();
    // Start at 100% power
    OCR0B = 255;
    while (1)
    {
    }
}
```

AVRDUDE Output

```
Output
Show output from: USBASP M328P
Error: cannot set sck period; please check for usbasp firmware update
Error: cannot set sck period; please check for usbasp firmware update
Reading 380 bytes for flash from input file btrMotorController.hex
Writing 380 bytes to flash
Writing | ################################################## | 100% 0.29s
Reading | ################################################## | 100% 0.20s
380 bytes of flash verified

Avrdude done.  Thank you.
```

Video

https://youtube.com/shorts/LI2515mfMRE?si=N34tyXfz-mZ-McJo